

Boolean algebra and flags

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Introduction

- programs contain logical decisions
- these are the "arguments" of if and while etc.
 - if a < 5
 - while t == 0
- these can be complicated (in pseudocode)

```
if (NOT network connected) OR (primary disk full AND no second disk) then
    what?
end if
```
- conditions can change while looping
- we still use math examples before we get to lists
- IB syllabus: B2.3.2, B2.3.3

Boolean values and negation

- Boolean is either True or False
- in Python, this variable type is bool
- *Boolean operators* produce new Booleans
- unary negation operator; Python not

a	not a
True	False
False	True

- exploration

Logical and, logical or

- binary operators for combining two logical values
- Python and

a	b	a and b
False	False	False
False	True	False
True	False	False
True	True	True

- Python or

a	b	a or b
False	False	False
False	True	True
True	False	True
True	True	True

- exploration

Logical operator precedence in Python

- operator precedence dictates following order

1. comparison operators, such as `<` and `!=`
2. logical negation `not`
3. `and`
4. `or`

- this means parentheses can be left out in some cases

- example:

```
((a <= b) and (not (b <= c))) or (a < c)
```

is equivalent to

```
a <= b and not b <= c or a < c
```

- however, use parentheses for clarity

Example 1

Fill in the missing Boolean values in the following *truth table*.

A	B	NOT A	NOT B	(A AND (NOT B))	((NOT A) AND B)	(A AND (NOT B)) OR ((NOT A) AND B)
F	F	T	T	F	F	F
F	T	T	F	F	T	T
T	F	F	T	T	F	T
T	T	F	F	T	F	T

Example 2

According to Wikipedia, in the Gregorian calendar a year is a leap year if it

is an integer multiple of 4 (except for years evenly divisible by 100, which are not leap years unless evenly divisible by 400)

Write a program which reports whether a year is a leap year. Test your code with years 2000, 2002, 2004 and 2100. Use

1. a Boolean variable and multiple if-statements
2. a single if-statement.

Boolean flag

- Boolean variable indicating whether a condition is met
- useful, among other things
 - for handling different (long) cases
 - for noticing when a loop can be escaped
 - especially in escaping from nested loops (later)

Input value validation using walrus operator

- *walrus operator* `:=` is assignment operator returning value
- used e.g. to concisely obtain and validate user input value in following style: $\hookrightarrow$

```
# note continuation of long line below by embedding logical statement
# within parentheses
while ((response := input('yes or no: ')) != 'yes'
      and response != 'no'):
    print('please answer yes or no')

print(response)
```

(we can do this even more concisely later)

- user input *type* validation is covered later

Example 3

James sleeps late if it is not a weekday ("arkipäivä") or if James is on vacation. Write a program which is given as input both

- the day of the week: integer 1–7 coding for Mon–Sun
- a yes/no-response indicating whether James is on vacation,

and outputs True if James sleeps late and False otherwise.

Homework 1

Jane should use sunscreen ("aurinkovoide") if the sun is shining and if current time is at least 10 and at most 16 (integer 0–23). Write a program which asks whether the sun is shining (yes/no-response) and asks for current time and outputs

Please use sunscreen.

if necessary.

Homework 2

Write a program which is given as input two positive integers and which outputs True if the two integers have the same last digit ("sama viimeinen numero") and outputs False otherwise.

Homework 3

Write a program which prompts the user for a positive integer s and finds the smallest positive integer value of n for which $n^3 - 10n^2 > s$. The program must print the found value of n and the corresponding value of the left hand side of the inequality.

$$n^3 - 10n^2 > s$$

$$n^2(n - 10) > s$$

$$n = 1$$

$$n = 0 \text{ or } 1$$

$$n > 10$$

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Homework 4

Write a program which takes as input two positive integers a and b and prints the smallest positive integer that is divisible by both a and b .

找公倍数

找最小公倍数

→ 找

$$\begin{array}{r} 20 \\ 2 \overline{) 20} \\ \underline{40} \\ 40 \end{array}$$

$$\begin{array}{r} 36 \\ 2 \overline{) 36} \\ \underline{72} \\ 72 \end{array}$$

$$2 \times 5 \times 2 \times 2 \times 2 = 10 \times 8 = 80$$

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Homework 5

Logical NAND statement is defined by

$$A \text{ NAND } B = \text{NOT } (A \text{ AND } B)$$

$$T \text{ NAND } T = \text{NOT } (T \text{ AND } T) = F$$

T	NAND	T	=	NOT	(T AND T)	=	F
T		F					T
F		T					T
F		F					T

1. Construct the truth table for NAND.

2. Construct the truth table for the statement

$$A \text{ NAND } A = \text{NOT } (A \text{ AND } A) = \text{NOT } A$$

Which more familiar logical statement does this correspond to?

3. Construct the truth table for the statement

$$(A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B) = A \text{ AND } B$$

Which more familiar logical statement does this correspond to?

Homework 6

A *prime (number)* is a positive integer greater than 1 that is divisible only by 1 and the prime number itself. Write a program which tests whether a given positive integer is a prime.

Homework 7 $(a \text{ AND } b) \text{ OR } (\text{NOT } a \text{ AND } \text{NOT } b)$
OR NOT AND.

Deduce a Boolean statement that corresponds to the truth table below.

~~1~~ $\neg (a \vee b)$

a	b	statement
False	False	True
False	True	False
True	False	False
True	True	True

~~2~~ $\neg a \vee b$

a	b	statement
False	False	True
False	True	True
True	False	False
True	True	True

~~3~~ $a \vee \neg b$

a	b	statement
False	False	True
False	True	False
True	False	True
True	True	True

Demonstrate that your statement is correct by constructing its truth table. (4) (10) (10) (10) (10)

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Homework 8

Write a program that prints

1. primes smaller than 100
2. 100 smallest primes.

Homework 9

A positive integer $n > 1$ is a perfect number if it is a sum of its proper divisors (integers smaller than n which are factors of n). An example of a perfect number is

$$6 = 1 + 2 + 3.$$

Write a program to find perfect numbers smaller than 10000.

28 1 7 14 4 1

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Homework 10

1. Find the ratio of the areas of the regions

$$R_1 : -1 \leq x \leq 1 \text{ and } -1 \leq y \leq 1$$

$$R_2 : x^2 + y^2 \leq 1.$$

2. The call `random.uniform(-1, 1)` gives a random value in the range $] -1, 1[$. Take 10000 random values of $-1 < x < 1$ and $-1 < y < 1$ and find the proportion of points landing in R_2 to all points.

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